Isomo Pulse Starfleet Monitor - Project Tracker

Document date: April 26, 2026

1. Project Summary

Starfleet Monitor is being built to give Isomo a reliable operational view of school connectivity. The current push is focused on making the laptop agent production-ready through Intune, improving backend site inference from Starlink identity, and making the dashboard reflect real site/device health.

2. Current Status

Workstream	Status	Notes
Backend API	In progress, deployable	Railway backend is healthy; recent work added Starlink identity site resolution and agent token generation
Neon database	In progress, deployable	Migrations include Starlink UUID/site master sheet fields and usage hardening
Windows agent	In progress, test-ready	Agent version 1.2.0 supports gRPC-web fallback, Starlink ID extraction, offline queue, and Intune marker
Intune deployment	In progress, VM test	Moving from split Platform Script/remediation to one self-contained Remediation package
Dashboard	In progress	Sites, laptops, usage, and detail views are evolving
Mobile app	Early/in progress	React Native app exists; field/admin workflows still need validation
Documentation	In progress	Root README, requirements, tracker, Intune setup, and site detection docs now exist

3. Recently Completed

Date	Item	Result
2026-04-26	Added backend Starlink identity matching	Backend can normalize `ut31...` to `31...` and match `sites.starlink_uuid`
2026-04-26	Added admin agent token route	Admin can create site-scoped agent tokens through `/api/agent-tokens`
2026-04-26	Added Intune remediation builder	`build-intune-remediation.mjs` generates a self-contained upload script
2026-04-26	Added Intune install marker	Agent install writes `install_source=intune_remediation` and `agent_version=1.2.0`
2026-04-26	Tightened detection script	Detection now forces remediation for old/manual installs

Date	Item	Result
2026-04-26	Confirmed VM old install symptoms	Old install lacked marker and showed `401 Unauthorized` ingest failures

4. Next Milestones

Milestone	Target outcome	Owner	Status
M1: Generate real agent token	Site-scoped token for test site 7 is created from production backend	Admin/Codex	Ready
M2: Upload new Intune remediation	Detection uses `packages/agent/detection.ps1`; remediation uses generated `dist/intune/remediation.ps1` with real token	Admin	Ready
M3: Validate VM remediation	VM has install marker, version 1.2.0, scheduled task, fresh heartbeat, queue 0, no `401`	Admin/Codex	Pending
M4: Confirm platform ingest	Backend receives heartbeat, health, latency, usage, and agent-health from VM	Admin/Codex	Pending
M5: Confirm Starlink site inference	Backend maps Starlink UUID to the correct real site when GPS is unavailable	Admin/Codex	Pending
M6: Expand pilot	Assign remediation to a small school/device group after VM validation	Admin/Ops	Pending
M7: Production rollout	Deploy to all target managed laptops with monitoring and rollback plan	Admin/Ops	Pending

5. Workstream Tracker

5.1 Backend And Database

Task	Priority	Status	Notes
Health endpoint verifies DB	Must	Done	`/health` reports API and DB status
Ingest endpoints accept agent payloads	Must	Done	Heartbeat, signal, health, latency, usage, agent-health
Idempotent queued payload replay	Must	Done	Payload metadata protects usage writes
Starlink UUID site lookup	Must	Done	Uses raw and normalized terminal identities
Agent token generation API	Must	Done	Admin route exists and returns site-scoped token
Master sheet site import/update	Must	In progress	Site ID, UUID, SN, location, district fields are modeled

Task	Priority	Status	Notes
Token rotation and revocation policy	Should	Not started	Needs operations decision

5.2 Windows Agent

Task	Priority	Status	Notes
Scheduled task every 5 minutes	Must	Done	`StarfleetPulse`
Built-in Starlink gRPC-web probe	Must	Done	Works without `grpcurl`
GPS byte-offset fallback	Must	Done	Decodes known diagnostic offsets
Starlink ID scanner	Must	Done	Sends `starlink_id` and `starlink_uuid`
Offline queue and replay	Must	Done	Queue stored in ProgramData
Agent health payload	Must	Done	Version, queue, last error, run ID
Intune install marker	Must	Done	Distinguishes new installs from legacy/manual installs
VM validation with new token	Must	Pending	Current VM evidence was old token/manual install

5.3 Intune Deployment

Task	Priority	Status	Notes
Single remediation package pattern	Must	Done	Do not split with Platform Script
Detection script forces remediation for old installs	Must	Done	Requires Intune marker and version
Self-contained remediation upload script	Must	Done	Generated under `dist/intune`
Upload to Intune test VM group	Must	Pending	Needs real token pasted into generated script
Verify Intune device status	Must	Pending	Expect detection with issues, remediation fixed, then without issues
Disable legacy Platform Script during test	Should	Pending	Avoid duplicate installers touching same files

5.4 Dashboard And Mobile

Task	Priority	Status	Notes
Sites overview	Must	In progress	Shows site inventory and metrics

Task	Priority	Status	Notes
Laptop list	Must	In progress	Tracks online/stale devices
Site detail	Must	In progress	Shows devices and latest signal
Usage chart	Should	In progress	New component exists in worktree
Alerts/site changes UI	Should	In progress	Backend events available
Mobile app field workflows	Should	Not validated	Needs test plan

6. Validation Checklist

6.1 Before Uploading To Intune

Check	Expected
Generated remediation has real agent token	`$ApiToken` is not `<PASTE_SITE_AGENT_JWT_HERE>`
Detection script is latest	Requires `install_source.json` and version `1.2.0`
Legacy Platform Script is unassigned	Only the remediation controls install/update
Assignment is scoped to VM test group	No broad production rollout yet

6.2 On The VM After Intune Runs

Run:

```
$dir = "C:\ProgramData\Starfleet"
Test-Path "$dir\install_source.json"
Get-Content "$dir\install_source.json" | ConvertFrom-Json
Select-String -Path "$dir\agent.log" -Pattern "Agent starting version|install_source|401|Cycle complete" | Select-
        Object -Last 30
Get-Content "$dir\last_heartbeat.txt" -ErrorAction SilentlyContinue
(Get-ChildItem "$dir\queue" -Filter "*.json" -ErrorAction SilentlyContinue).Count
```

Expected:

Evidence	Good result
Install marker	`source = intune_remediation`
Agent version	`agent_version = 1.2.0`
Log startup	`Agent starting version 1.2.0 ... install_source=intune_remediation`
Auth	No `401 Unauthorized` ingest failures
Queue	`0` after successful replay
Heartbeat	`last_heartbeat.txt` updated after Intune run

6.3 In The Platform

Check	Expected
Device last seen	VM/laptop updates within expected interval
Agent health	Version 1.2.0 and queue depth visible
Site assignment	Real site is resolved from Starlink UUID/GPS or fallback
Usage	Daily usage appears without duplicate spikes
Latency	Recent laptop-side latency appears

7. Risks And Blockers

Risk	Severity	Current mitigation
Wrong token type used in Intune	High	Use `/api/agent-tokens` to generate site-scoped token; verify no `401`
Intune old Platform Script still assigned	Medium	Unassign during test so remediation is the only installer
Starlink local API unreachable	Medium	Agent still sends laptop heartbeat/health/usage; next cycle can recover
Starlink GPS disabled by update/config	Medium	Backend uses Starlink UUID inventory as primary fallback
Generated token script committed	High	`dist/` ignored; review `git status` before commits
Incomplete site inventory	Medium	Continue importing Starlink UUID/SN/location/site records

8. Decisions Needed

Decision	Why it matters	Suggested next step
Agent token lifetime	Controls security and operational maintenance	Start with 365 days for pilot, define rotation later
Production rollout grouping	Controls blast radius	Pilot with VM, then one small school group, then expand
Dashboard domain and CORS	Prevents auth/API failures	Confirm final Vercel URL and backend `ALLOWED_ORIGINS`
Notification recipients	Determines who gets site move alerts	Define admin/ops recipients and acknowledgement workflow
Site master sheet ownership	Prevents stale UUID/location data	Assign one source-of-truth owner

9. Immediate Next Actions

1. Generate a production site-scoped agent token for site 7.
2. Paste it into `dist/intune/remediation.ps1`.

3. Upload latest detection and remediation scripts to the Intune remediation.
4. Sync the VM and wait for remediation status.
5. Run the VM validation commands.
6. Confirm backend/platform shows fresh ingest without `401`.
7. If the VM passes, pilot a small real laptop group.

10. Definition Of Done For Current Phase

This phase is complete when:

Criterion	Target
VM install	VM shows Intune marker and agent version 1.2.0
VM ingest	Heartbeat, health, latency, usage, and agent-health ingest without auth failures
Starlink mapping	Starlink UUID resolves the correct site when GPS is unavailable
Dashboard	Operators can see VM/laptop health and site assignment in the platform
Runbook	Intune setup and validation steps are documented and repeatable